DRank_herald

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1 DRank Keyword Extraction (Herald Dataset)

Graph-based DRank implementation for the same dataset used in HRank. This notebook ranks candidate keywords using a token co-occurrence graph and PageRank-style scoring.

[1]:
```
import re
import gzip
import zlib
import urllib.request
from collections import Counter, defaultdict

from bs4 import BeautifulSoup, Comment
import nltk
from nltk.corpus import stopwords, wordnet
from nltk import wordpunct_tokenize, pos_tag
from nltk.stem import PorterStemmer, WordNetLemmatizer

for pkg in ["stopwords", "wordnet", "omw-1.4", "averaged_perceptron_tagger"]:
    try:
        if pkg == "averaged_perceptron_tagger":
            nltk.data.find("taggers/averaged_perceptron_tagger")
        else:
            nltk.data.find(f"corpora/{pkg}")
    except LookupError:
        nltk.download(pkg, quiet=True)

COMMON_NOISE_WORDS = set(
    """
january debt est dec big than who use jun jan feb mar apr may jul agust dec oct␣
↪nov sep dec
product continue one two three four five please thanks find helpful week job␣
↪experience women girl
apology read show eve knowledge benefit appointment street way staff salon␣
↪discount gift cost thing
world close party love letters rewards offers special close page week dollars␣
↪voucher gifts vouchers
welcome therefore march nights need name pleasure show sisters thank menu today␣
↪always time needs
```

```
welcome march february april may june jully aguast september october november␣
↪december day year
month minute second secodns
""".split()
)

SPECIAL_CHARS_RE = re.compile(
    r"[ \~\!\@\#\\\$\%\^\&\*\(\)\_\+\=\\\|\{\}\[\]\:\;\'\"\<\>\,\/\.\-]"
)

stemmer = PorterStemmer()
lemmatizer = WordNetLemmatizer()
POS_MAP = {"J": wordnet.ADJ, "V": wordnet.VERB, "N": wordnet.NOUN, "R": wordnet.
↪ADV}

def _is_visible_text(element) -> bool:
    if element.parent.name in ["html", "style", "script", "head", "[document]",␣
↪"img"]:
        return False
    if isinstance(element, Comment):
        return False
    return True

def _extract_visible_text_from_html(html: bytes) -> str:
    soup = BeautifulSoup(html, "lxml")
    texts = soup.find_all(string=True)
    visible_texts = filter(_is_visible_text, texts)
    return " ".join(t.strip() for t in visible_texts if t and t.strip())

def _normalize_whitespace(text: str) -> str:
    lines = (line.strip() for line in text.splitlines())
    chunks = (phrase.strip() for line in lines for phrase in line.split(" "))
    return "\n".join(chunk for chunk in chunks if chunk)

def _decode_response_body(html: bytes, encoding: str) -> bytes:
    enc = (encoding or "").lower()
    if enc == "gzip":
        try:
            return gzip.decompress(html)
        except Exception:
            return html
    if enc == "deflate":
        try:
            return zlib.decompress(html)
        except Exception:
            try:
                return zlib.decompress(html, -zlib.MAX_WBITS)
```

```
            except Exception:
                return html
    return html

def _fetch_page(u: str):
    req = urllib.request.Request(
        u,
        headers={
            "User-Agent": "Mozilla/5.0",
            "Accept": "text/html,application/xhtml+xml,application/xml;q=0.9,*/
 ↪*;q=0.8",
            "Accept-Language": "en-US,en;q=0.9",
            "Connection": "keep-alive",
        },
    )
    with urllib.request.urlopen(req, timeout=20) as resp:
        html = resp.read()
        html = _decode_response_body(html, resp.headers.get("Content-Encoding",␣
 ↪""))
        clean_text =␣
 ↪_normalize_whitespace(_extract_visible_text_from_html(html))
        return clean_text

def _calculate_language_scores(text: str) -> dict:
    ratios = {}
    tokens = wordpunct_tokenize(text)
    words = [w.lower() for w in tokens]
    words_set = set(words)
    for lang in stopwords.fileids():
        try:
            stop_set = set(stopwords.words(lang))
            ratios[lang] = len(words_set.intersection(stop_set))
        except Exception:
            continue
    return ratios

def _detect_language_and_stopwords(text: str):
    ratios = _calculate_language_scores(text)
    if not ratios:
        return "english", set(stopwords.words("english"))
    detected_lang = max(ratios, key=ratios.get)
    try:
        sw = set(stopwords.words(detected_lang))
    except Exception:
        detected_lang, sw = "english", set(stopwords.words("english"))
    return detected_lang, sw
```

```python
def _stem_and_lemmatize_tokens(tokens):
    if not tokens:
        return []
    tagged = pos_tag(tokens)
    out = []
    for word, tag in tagged:
        wn_tag = POS_MAP.get(tag[0], wordnet.NOUN)
        lemma = lemmatizer.lemmatize(word, wn_tag)
        stem = stemmer.stem(lemma)
        out.append(stem)
    return out

def _filter_candidate_pos(tokens):
    tagged = pos_tag(tokens)
    return [w for w, tag in tagged if tag.startswith("NN") or tag.
 ↪startswith("JJ")]

def normalize_keywords(keywords):
    cleaned = [SPECIAL_CHARS_RE.sub("", w.lower().strip()) for w in keywords if␣
 ↪w]
    cleaned = [w for w in cleaned if w and len(w) > 1 and w.isascii() and not w.
 ↪isdigit()]
    return _stem_and_lemmatize_tokens(cleaned)

def _clean_text_to_words(text: str, stopword_list: set) -> list:
    tokens = [SPECIAL_CHARS_RE.sub("", w.lower().strip()) for w in␣
 ↪wordpunct_tokenize(text)]
    tokens = [
        t for t in tokens
        if t and len(t) > 1 and t.isascii() and not t.isdigit()
        and t not in stopword_list and t not in COMMON_NOISE_WORDS
    ]
    tokens = _filter_candidate_pos(tokens)
    return _stem_and_lemmatize_tokens(tokens)

def Get_Prc_Rcl_Fscr_input_GT_and_Keywords_List(ground_truth, keywords):
    gt_set = set(ground_truth)
    kw_set = set(keywords)
    matches = gt_set.intersection(kw_set)
    gt_count, kw_count, match_count = len(gt_set), len(kw_set), len(matches)
    if gt_count == 0 or kw_count == 0:
        return (0, 0, 0)
    precision = match_count / kw_count
    recall = match_count / gt_count
    f_score = (2 * precision * recall) / (precision + recall) if (precision +␣
 ↪recall) > 0 else 0
    return (precision, recall, f_score)
```

```python
def read_url(url):
    with urllib.request.urlopen(url) as f:
        return f.read().decode("utf-8-sig", errors="ignore").strip()

def load_herald_case(index):
    base = "https://cs.uef.fi/~himat/WebRank/dataset_12/dataset_12/herald"
    url_text = read_url(f"{base}/{index}/URL.txt")
    gt_text = read_url(f"{base}/{index}/GT.txt")
    return url_text, gt_text.split()
```

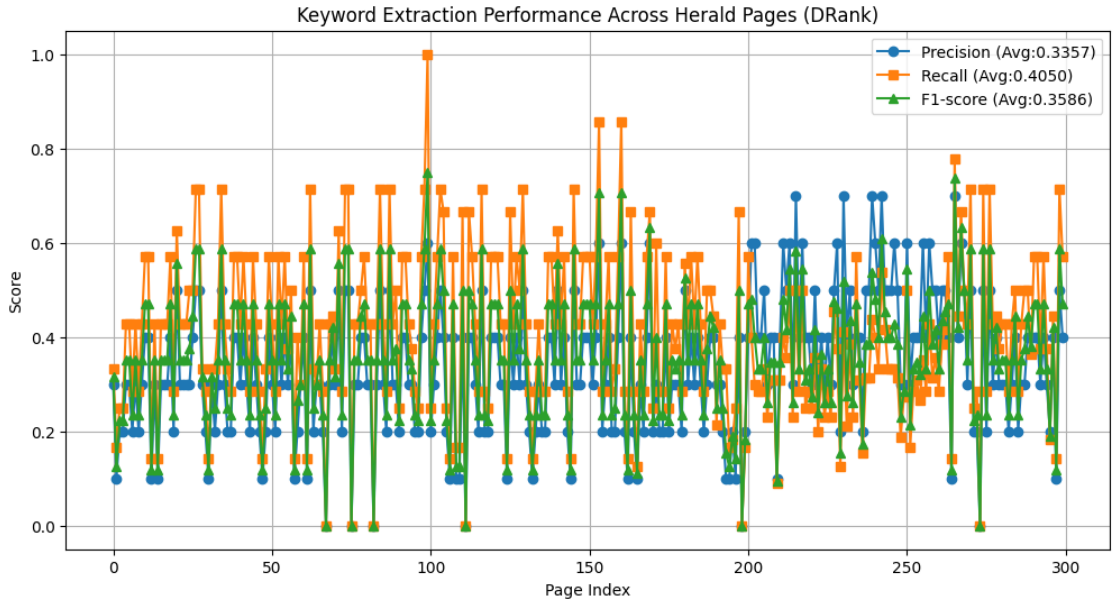
[2]:
```python
def _build_cooccurrence_graph(tokens, window_size=4):
    graph = defaultdict(lambda: defaultdict(float))
    n = len(tokens)
    for i in range(n):
        wi = tokens[i]
        end = min(i + window_size, n)
        for j in range(i + 1, end):
            wj = tokens[j]
            if wi == wj:
                continue
            dist = j - i
            weight = 1.0 / dist
            graph[wi][wj] += weight
            graph[wj][wi] += weight
    return graph

def _run_pagerank(graph, damping=0.85, max_iter=100, tol=1e-6):
    nodes = list(graph.keys())
    if not nodes:
        return {}

    n = len(nodes)
    score = {node: 1.0 / n for node in nodes}
    out_weight_sum = {node: sum(graph[node].values()) for node in nodes}

    for _ in range(max_iter):
        new_score = {node: (1.0 - damping) / n for node in nodes}

        dangling_mass = sum(score[node] for node in nodes if␣
 ↪out_weight_sum[node] == 0.0)
        dangling_share = damping * dangling_mass / n
        for node in nodes:
            new_score[node] += dangling_share

        for src in nodes:
            if out_weight_sum[src] == 0.0:
```

```
                continue
            for dst, w in graph[src].items():
                new_score[dst] += damping * score[src] * (w /␣
↪out_weight_sum[src])

        diff = sum(abs(new_score[node] - score[node]) for node in nodes)
        score = new_score
        if diff < tol:
            break

    return score

def get_top_keywords_drank(url: str, k: int = 10, window_size: int = 4,␣
↪return_details: bool = False):
    clean_text = _fetch_page(url)
    _, sw = _detect_language_and_stopwords(clean_text)
    tokens = _clean_text_to_words(clean_text, sw)
    if not tokens:
        return []

    graph = _build_cooccurrence_graph(tokens, window_size=window_size)
    drank_scores = _run_pagerank(graph)

    tf = Counter(tokens)
    final_scores = {}
    for w, c in tf.items():
        final_scores[w] = c * drank_scores.get(w, 0.0)

    top = sorted(final_scores.items(), key=lambda kv: kv[1], reverse=True)[:k]
    if return_details:
        return [(w, tf[w], s) for w, s in top]
    return [w for w, _ in top]
```

[3]:
```
def run_single_case_drank(index=0, k=10, window_size=6):
    url, gt_keywords = load_herald_case(str(index))
    gt_keywords_norm = normalize_keywords(gt_keywords)
    found_keywords = get_top_keywords_drank(
        url,
        k=k,
        window_size=window_size
    )
    p, r, f = Get_Prc_Rcl_Fscr_input_GT_and_Keywords_List(gt_keywords_norm,␣
↪found_keywords)
    return {
        "index": index,
        "url": url,
        "ground_truth": gt_keywords,
```

```
        "ground_truth_normalized": gt_keywords_norm,
        "predicted": found_keywords,
        "precision": round(p, 4),
        "recall": round(r, 4),
        "f_score": round(f, 4),
    }

def run_evaluation_drank(total_webpages=10, k=10, window_size=6):
    p_sum, r_sum, f_sum = 0.0, 0.0, 0.0
    valid = 0
    for i in range(total_webpages):
        try:
            url, gt_keywords = load_herald_case(str(i))
            gt_keywords_norm = normalize_keywords(gt_keywords)
            found_keywords = get_top_keywords_drank(
                url,
                k=k,
                window_size=window_size
            )
            p, r, f =␣
 ↪Get_Prc_Rcl_Fscr_input_GT_and_Keywords_List(gt_keywords_norm, found_keywords)
            p_sum += p
            r_sum += r
            f_sum += f
            valid += 1
        except Exception:
            continue

    if valid == 0:
        return {
            "average_precision": 0,
            "average_recall": 0,
            "average_f_score": 0,
            "total_webpages": total_webpages,
            "evaluated_webpages": 0,
        }

    return {
        "average_precision": round(p_sum / valid, 4),
        "average_recall": round(r_sum / valid, 4),
        "average_f_score": round(f_sum / valid, 4),
        "total_webpages": total_webpages,
        "evaluated_webpages": valid,
    }
```

[4]:
```
# Visualize and compare performance for DRank (Herald)
import matplotlib.pyplot as plt
```

```python
import os
import pandas as pd

results = []
for i in range(300):
    case = run_single_case_drank(index=i, k=10, window_size=4)
    results.append(case)
dataset_precisions = [r['precision'] for r in results]
dataset_recalls = [r['recall'] for r in results]
dataset_f1s = [r['f_score'] for r in results]

plt.figure(figsize=(12, 6))
plt.plot(dataset_precisions, label='Precision (Avg:{0:.4f})'.
 ↪format(sum(dataset_precisions)/len(dataset_precisions)), marker='o')
plt.plot(dataset_recalls, label='Recall (Avg:{0:.4f})'.
 ↪format(sum(dataset_recalls)/len(dataset_recalls)), marker='s')
plt.plot(dataset_f1s, label='F1-score (Avg:{0:.4f})'.format(sum(dataset_f1s)/
 ↪len(dataset_f1s)), marker='^')
plt.xlabel('Page Index')
plt.ylabel('Score')
plt.title('Keyword Extraction Performance Across Herald Pages (DRank)')
plt.legend()
plt.grid(True)

# Export graph as image
results_dir = '../Results'
os.makedirs(results_dir, exist_ok=True)
image_path = os.path.join(results_dir, 'DRank_Herald.png')
plt.savefig(image_path)
plt.show()

# Update results.csv
csv_path = os.path.join(results_dir, 'results.csv')
method = 'DRank_Herald'
avg_precision = round(sum(dataset_precisions)/len(dataset_precisions), 4)
avg_recall = round(sum(dataset_recalls)/len(dataset_recalls), 4)
avg_f1 = round(sum(dataset_f1s)/len(dataset_f1s), 4)

row = {'DataSet': method, 'Precision': avg_precision, 'Recall': avg_recall,␣
 ↪'F1-score': avg_f1, 'Evaluated Pages': len(results)}

if os.path.exists(csv_path) and os.path.getsize(csv_path) > 0:
    df = pd.read_csv(csv_path)
    df = df[df['DataSet'] != method]
    df = pd.concat([df, pd.DataFrame([row])], ignore_index=True)
else:
    df = pd.DataFrame([row])
```

```python
df.to_csv(csv_path, index=False)
```